Practice Exercise 3 (additional)

K-Means Image Segmentation Exercise

Problem: Segment a given image into K clusters using the K-means algorithm

Dataset: You can use any image of your choice. For example, a natural image like a landscape or a portrait.

Task:

1. Load the Image:

• Read the image using OpenCV or PIL.

• Convert the image to a suitable color space (e.g., RGB, HSV , or LAB).

2. Preprocess the Image:

• Resize the image to a smaller size for faster processing.

• Flatten the image into a 2D array of pixels, where each pixel is represented as a feature vector (e.g., RGB values).

3. Apply K-Means Clustering:

• Initialize K random centroids, each representing the mean color of a cluster.

• Assign each pixel to the nearest centroid based on Euclidean distance.

• Update the centroids as the mean of the pixels assigned to each cluster.

• Repeat the assignment and update steps until convergence.

4. Segment the Image:

• Replace each pixel with the color of its assigned cluster centroid.

• Reshape the segmented image back to its original dimensions.

5. Visualize the Results:

• Display the original and segmented images side-by-**side**